

# Reading `TaylorScan.v`, `TaylorReal.v` and `TaylorLink.v`

Zimmermann's hard-to-round search in Rocq, for a beginner

Three files in `code/APaul/rocq/` state the search of `doc/htr.md` and prove that it misses no hard-to-round case, assuming the evaluation of `exp` and its Taylor bound:

- `TaylorScan.v`: the search over the integers, and that it returns every  $j$  where the integer polynomial is close to a multiple of  $M$ ;
- `TaylorReal.v`: the step from `exp` to that integer polynomial, on the real numbers;
- `TaylorLink.v`: the two put together, in the final theorem `scan_exp`.

This note explains the idea, then the files. The three files are in the annexes. None has an **Admitted**.

## 1 The problem

A double  $x$  is **hard to round** when  $\exp x$  is extremely close to a point where rounding changes its mind. Measured in a unit  $v$  (half the distance between two doubles near  $\exp x$ ), the rounding points are the integers, so  $x$  is hard to round at level  $m$  when  $\exp(x)/v$  is within  $2^{-m}$  of an integer. We want every such  $x$  in a given range, without evaluating  $\exp$  at each of the billions of doubles.

## 2 Zimmermann's idea in four steps

1. **Cut the range** into subranges of  $n$  consecutive doubles  $x_0, x_0 + u, \dots, x_0 + (n-1)u$ , where  $u$  is the distance between two doubles. We number them  $j = 0, \dots, n-1$ .
2. **Replace  $\exp$  by a polynomial in  $j$** . On a subrange,  $\exp(x_0 + ju)/v$  is very close to a polynomial of small degree in  $j$  (its Taylor polynomial). Multiply by a large power of 2,  $M = 2^{64\ell}$ , and round the coefficients: we get a polynomial with **integer** coefficients,

$$P(j) = A_0 + A_1j + \dots + A_{k-1}j^{k-1}.$$

Only the part of  $P(j)$  modulo  $M$  matters, because only the distance to an integer matters.

3. **Test each  $j$** .  $x_0 + ju$  is a **candidate** when  $P(j)$  is within a margin  $E$  of a multiple of  $M$ .  $E$  is chosen to cover everything we neglected: the target  $2^{-m}$ , the Taylor remainder, and the rounding of the coefficients.
4. **Compute  $P(0), P(1), P(2), \dots$  with additions only**, by the table of differences (next section). This is what makes the search fast.

**The test in one formula.** “ $b$  is within  $E$  of a multiple of  $M$ ” is checked as

$$(b + E) \bmod M \leq 2E.$$

Example with  $M = 100$  and  $E = 5$ :  $b = 197$  is within 3 of 200, and  $(197 + 5) \bmod 100 = 2 \leq 10$ : a candidate.  $b = 150$  gives  $(150 + 5) \bmod 100 = 55 > 10$ : not a candidate. Adding  $E$  shifts the interval  $[-E, E]$  around a multiple of  $M$  to  $[0, 2E]$ , which a single comparison can test.

## 3 The table of differences

The **difference** of a sequence  $f$  at  $j$  is  $\Delta f(j) = f(j+1) - f(j)$ . For a polynomial of degree  $d$ , taking  $d$  differences leaves a constant, and  $d+1$  differences leave 0. The **table** at  $j$  is the list  $f(j), \Delta f(j), \Delta^2 f(j), \dots$

Example with  $f(j) = j^2$ : the table at  $j = 0$  is  $[0, 1, 2]$  (the value 0, then  $1 - 0 = 1$ , then  $(4 - 1) - (1 - 0) = 2$ ). **One step** adds each row to the one below it:

| $j$ | $f(j)$ | $\Delta f(j)$ | $\Delta^2 f(j)$ |
|-----|--------|---------------|-----------------|
|-----|--------|---------------|-----------------|

|   |             |             |   |
|---|-------------|-------------|---|
| 0 | 0           | 1           | 2 |
| 1 | $0 + 1 = 1$ | $1 + 2 = 3$ | 2 |
| 2 | $1 + 3 = 4$ | $3 + 2 = 5$ | 2 |
| 3 | $4 + 5 = 9$ | $5 + 2 = 7$ | 2 |

The first column gives 0, 1, 4, 9: the squares, with additions only. This is **Shift.v**, already proved:

- **dtab d f j** is the table of  $f$  at  $j$ , with  $d + 1$  rows;
- **tstep** is one step;
- **tstepE**: if  $f$  has degree at most  $d$  (**degle d f**), one step turns the table at  $j$  into the table at  $j + 1$ .

## 4 The file, line by line

**Notations used.** **int** is the type of integers of the library **mathcomp**.  $x \text{ }^{*+} \text{ } n$  is  $x$  added  $n$  times, that is  $nx$ .  $(a \text{ } \% M) \% Z$  is  $a \bmod M$ , between 0 and  $M - 1$ .  $|x|$  is the absolute value. **iota 0 n** is the list 0, 1, ...,  $n - 1$ , and **[seq j <- s | p j]** keeps the elements  $j$  of  $s$  for which  $p \ j$  holds.  $j \setminus \text{in } s$  means  $j$  is in the list  $s$ .  $(j < n) \% N$  is a comparison of natural numbers.

**The data.** One subrange is described by four values:

**Variables** (A : seq int) (M E : int) (n : nat).

the coefficients  $A_0, \dots, A_{k-1}$  as a list, the modulus  $M$ , the margin  $E$ , and the number  $n$  of doubles.

**The polynomial.**

**Definition** P : nat -> int := hpoly A.

**hpoly** (from **Shift.v**) is the polynomial with coefficients **A** in Horner form:  $A_0 + j(A_1 + j(A_2 + \dots))$ . Its degree is at most the length of **A**:

**Definition** dg := size A.

**Lemma** degleP : degle dg P.

**The test**, the formula of section 2:

**Definition** hit (b : int) : bool := ((b + E) \% M) \% Z <= E \*+ 2.

**The walk**, step 6 of the note. It keeps the current table **t** and the current index **j**, tests the first row, then takes one step; **c** counts the doubles left:

```
Fixpoint walk (j : nat) (t : seq int) (c : nat) : seq nat :=
  if c is c'.+1 then
    let rest := walk j.+1 (tstep t) c' in
    if hit (nth 0 t 0) then j :: rest else rest
  else [::].
```

**nth 0 t 0** is the first row, the current value  $P(j)$ . The result is the list of candidates.

**The search** starts the walk from the table of  $P$  at 0:

**Definition** scan : seq nat := walk 0 (dtab dg P 0) n.

## 5 What is proved

**The walk computes the right thing.**

**Lemma** scanE : scan = [seq j <- iota 0 n | hit (P j)].

The walk returns exactly the  $j < n$  whose  $P(j)$  passes the test: the fast computation with additions gives the same answer as evaluating  $P$  at every  $j$ . The proof (**walkE**) goes by induction on the number

of doubles left; at each step `tstepE` says the table after one step is the table at the next index, whose first row is  $P(j+1)$ .

**The test catches every  $b$  close to a multiple of  $M$ .**

Lemma `hitP` (b w : int) :  $E * 2 < M \rightarrow \neg |b - M * w| \leq E \rightarrow \text{hit } b$ .

If  $|b - Mw| \leq E$ , then  $b - Mw + E$  is between 0 and  $2E$ , which is less than  $M$ , so it is  $(b + E) \bmod M$ . The condition  $2E < M$  makes sure the interval  $[0, 2E]$  does not wrap around.

**The main theorem.**

Theorem `scan_complete` j (w : int) :  
 $E * 2 < M \rightarrow (j < n) \% N \rightarrow \neg |P \ j - M * w| \leq E \rightarrow j \ \backslash \text{in } \text{scan}$ .

If  $P(j)$  is within  $E$  of a multiple of  $M$ , then  $j$  is a candidate. It is `scanE` and `hitP` put together. The file has no `Admitted`.

## 6 From exp to the scan: `TaylorReal.v`

`scan_complete` talks about  $P$ , not about  $\exp$ . `TaylorReal.v` makes the step between them: **if  $x_0 + ju$  is hard to round, then  $P(j)$  is within  $E$  of a multiple of  $M$** . It holds under the three assumptions of the note. With  $a_i = \exp(x_0)u^i/(i!v)$  the exact coefficients and  $\rho$  a bound on the Taylor remainder:

- (H\_A) each  $A_i$  is within 1 of  $M$  times the fractional part of  $a_i$ : this is the evaluation of  $\exp$ , done once per subrange;
- (H\_T)  $\exp(x_0 + ju)/v$  is within  $\rho$  of  $a_0 + a_1j + \dots + a_{k-1}j^{k-1}$ ;
- (H\_E)  $E \geq M(2^{-m} + \rho) + (1 + n + \dots + n^{k-1})$ .

The file is plain Rocq on real numbers (the `Stdlib` library, no `mathcomp`). In it,  $\exp$  does not appear: the value  $\exp(x_0 + ju)/v$  is a real  $y$ , and  $2^{-m}$  a real  $\varepsilon$ . Two definitions:

- `sumR k f` is  $f(0) + \dots + f(k-1)$ ;
- `Pz A k j` is the integer  $A_0 + A_1j + \dots + A_{k-1}j^{k-1}$ .

The lemma:

Lemma `real_lemma` (k n j : nat) (M E : Z) (a : nat -> R) (A : nat -> Z)  
 (rho eps y : R) :  
 $(0 < M) \% Z \rightarrow (j \leq n) \% \text{nat} \rightarrow$   
 $(\text{forall } i, (i < k) \% \text{nat} \rightarrow \text{Rabs } (\text{IZR } (A \ i) - \text{IZR } M * \text{frac\_part } (a \ i)) < 1) \rightarrow$   
 $\text{Rabs } (y - \text{sumR } k \ (\text{fun } i \Rightarrow a \ i * \text{INR } j ^ i)) \leq \text{rho} \rightarrow$   
 $\text{IZR } M * (\text{eps} + \text{rho}) + \text{sumR } k \ (\text{fun } i \Rightarrow \text{INR } n ^ i) \leq \text{IZR } E \rightarrow$   
 $(\text{exists } z : Z, \text{Rabs } (y - \text{IZR } z) < \text{eps}) \rightarrow$   
 $\text{exists } w : Z, (Z.\text{abs } (\text{Pz } A \ k \ j - M * w) \leq E) \% Z$ .

`IZR` and `INR` turn an integer and a natural number into a real; `frac_part`  $r$  is  $r$  minus its integer part. The three middle hypotheses are (H\_A), (H\_T) and (H\_E); the last one says  $y$  is within  $\varepsilon$  of an integer  $z$ .

**The proof in four lines.** Write each  $a_i$  as its integer part plus its fractional part.

1.  $Ma_i j^i$  and  $M \text{frac}(a_i) j^i$  differ by  $M$  times an integer, because  $j^i$  is an integer. So  $M \times \sum a_i j^i$  is  $\sum M \text{frac}(a_i) j^i$  plus  $MK$  for an integer  $K$ .
2. Replacing each  $M \text{frac}(a_i)$  by  $A_i$  costs less than  $j^i \leq n^i$ , by (H\_A): the total error is at most  $1 + n + \dots + n^{k-1}$ .
3. Replacing  $\sum a_i j^i$  by  $y$  costs at most  $M\rho$ , by (H\_T); and  $y$  is within  $\varepsilon$  of  $z$ , so  $My$  is within  $M\varepsilon$  of  $Mz$ .
4. Take  $w = z - K$ :  $P(j) - Mw$  is the sum of the three errors, at most  $E$  by (H\_E).

The file proves it with five small lemmas on `sumR`: two sums can be added, multiplied by a constant, compared term by term, bounded by the triangle inequality, and `Pz` read in the reals is the real sum (`IZR_Pz`).

## 7 The final theorem: `TaylorLink.v`

`TaylorScan.v` is written with the `mathcomp` library, whose integers are the type `int` and whose polynomial is `hpoly A`; `TaylorReal.v` uses Rocq's `Z` and `Pz`. `TaylorLink.v` connects them:

- `hpoly_Pz`: `hpoly A j` and `Pz` compute the same integer (`Z_of_int` turns a `mathcomp int` into a `Z`);
- `scan_completeZ`: `scan_complete`, restated with `Z` and `Pz`;
- `scan_exp`: `real_lemma` followed by `scan_completeZ`.

```
Theorem scan_exp (A : seq int) (M E : int) (n j : nat) (a : nat -> R)
  (rho eps y : R) :
  Z.lt 0 (Z_of_int M) -> Z.lt (Z.mul 2 (Z_of_int E)) (Z_of_int M) ->
  (j < n)%N ->
  (H_A) -> (H_T) -> (H_E) ->
  (exists z : Z, Rlt (Rabs (Rminus y (IZR z))) eps) ->
  j \in scan A M E n.
```

(written here with  $(H\_A)$ ,  $(H\_T)$ ,  $(H\_E)$  for the three hypotheses; the file spells them out). In words: **if  $\exp(x_0 + ju)/v$  is within  $2^{-m}$  of an integer, then  $j$  is a candidate**. The search misses no hard-to-round case. The file writes the real and `Z` operations as functions (`Rle`, `Rplus`, `Z.add`, ...) because `mathcomp` takes over the infix notations  $\leq$ ,  $+$ , ... for its own types.

Print Assumptions `scan_exp` lists only the two axioms Rocq's real numbers are built on.

## 8 What is assumed

- $(H\_A)$ , the evaluation of `exp`: for `htr2_fix.c`, it is the bound of its Tang evaluator at 448 bits, not yet proved.
- $(H\_T)$ , the Taylor bound for `exp` with its remainder: a theorem of analysis, not yet proved here.
- Two details separate the files from the C program `htr2_fix.c`:
  - the C builds the first table in place from  $P(0), \dots, P(k-1)$  (steps 3 and 4 of the note), where `TaylorScan.v` takes `dtab` directly; and its table has one more row, which is 0;
  - the C reduces every number modulo  $M$  after each addition, where the file keeps exact integers and reduces only in the test; the two give the same test, because reduction modulo  $M$  commutes with addition.

## A Annex: `TaylorScan.v`

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(** * Zimmermann's search (doc/htr.md): the scan of one subrange
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```
On a subrange of [n] consecutive doubles [x0 + j u], the note replaces
[exp(x0 + j u) / v] by an integer polynomial in [j],
[P j = A_0 + A_1 j + ... + A_(k-1) j^(k-1)],
whose coefficients are the [A_i] of the note, and walks it with the
table of differences of [Shift.v]. A [j] is a candidate when
[(P j + E) mod M <= 2 E], with [M = beta^l].
```

```
This file states the algorithm over the integers and proves the
part that does not depend on [exp]:
```

- `[scanE]`: the walk returns exactly the  $[j < n]$  passing the test;
- `[scan_complete]`: every  $[j]$  with  $[P j]$  within  $[E]$  of a multiple of  $[M]$  is returned.

```

    The link to [exp] is in [TaylorReal.v] and [TaylorLink.v]. *)

From mathcomp Require Import all_ssreflect all_algebra.
From APaulRocq Require Import Shift.

Set Implicit Arguments.
Unset Strict Implicit.
Unset Printing Implicit Defensive.

Import GRing.Theory Num.Theory.
Local Open Scope ring_scope.

Section Scan.

(** The data of a subrange: the [A_i], least significant first, the
    modulus [M = beta^l], the window [E], and the number of doubles. *)
Variables (A : seq int) (M E : int) (n : nat).

(** The polynomial, in Horner form. *)
Definition P : nat -> int := hpoly A.

(** Its degree is at most [size A], so a table of [size A + 1] rows walks
    it exactly (the note uses [size A] rows: the last one here is [0]). *)
Definition dg := size A.

Lemma degP : degP dg P.
Proof. exact: degP_hpoly. Qed.

(** The test on the first row of the table: steps 5 and 6 of the note. *)
Definition hit (b : int) : bool := ((b + E) %% M)%Z <= E *+ 2.

(** The main loop, step 6: test the first row, then shift the table. *)
Fixpoint walk (j : nat) (t : seq int) (c : nat) : seq nat :=
  if c is c'.+1 then
    let rest := walk j.+1 (tstep t) c' in
    if hit (nth 0 t 0) then j :: rest else rest
  else [::].

(** The search: the table at [0], steps 3 and 4, then the walk. *)
Definition scan : seq nat := walk 0 (dtab dg P 0) n.

(** The walk returns exactly the [j] in [[i, i + c]] whose [P j] passes
    the test. *)
Lemma walkE i c : walk i (dtab dg P i) c = [seq j <- iota i c | hit (P j)].
Proof.
  elim: c i => [//|c IH] i /=.
  rewrite tstepE; last exact: degP.
  by rewrite IH.
Qed.

Lemma scanE : scan = [seq j <- iota 0 n | hit (P j)].
Proof. exact: walkE. Qed.

(** If [b] is within [E] of a multiple [M w] of [M], and [2 E < M], then
    [b] passes the test: [(b + E) mod M] is [b - M w + E], in [[0, 2 E]]. *)
Lemma hitP (b w : int) : E *+ 2 < M -> `|b - M * w| <= E -> hit b.

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Proof.
move=> EM; rewrite ler_norml => /andP[l r].
have lo : 0 <= b - M * w + E by rewrite -lerBlDr sub0r.
have hi : b - M * w + E < M.
  by apply: Order.POrderTheory.le_lt_trans EM; rewrite mulr2n lerD2r.
rewrite /hit.
have -> : b + E = w * M + (b - M * w + E).
  by rewrite [w * M]mulrC addrA [M * w + _]addrC subrK.
rewrite modzMDl modz_small ?lo ?hi //.
by rewrite mulr2n lerD2r.
Qed.

(** The completeness of the scan, with no reference to [exp]. *)
Theorem scan_complete j (w : int) :
  E *+ 2 < M -> (j < n)%N -> `|P j - M * w| <= E -> j \in scan.
Proof.
move=> EM jn Pw.
by rewrite scanE mem_filter (hitP EM Pw) /= mem_iota /= add0n.
Qed.

End Scan.

(** The step from [exp] to the hypothesis of [scan_complete] is
    [TaylorReal.real_lemma]; [TaylorLink.scan_exp] puts the two
    together. *)

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## B Annex: TaylorReal.v

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(** * Zimmermann's search (doc/htr.md): from the reals to the scan

[TaylorScan.scan_complete] says: if the integer polynomial [P] is
within [E] of a multiple of [M] at [j], then [j] is a candidate. This
file proves the step before it, on the reals: if [y] (standing for
[exp(x0 + j u) / v]) is within [eps] (standing for [2^-m]) of an
integer, then [P j] is within [E] of a multiple of [M], under the three
hypotheses of the note:
  (H_A) each [A_i] is within 1 of [M frac(a_i)];
  (H_T) [y] is within [rho] of [sum_i a_i j^i];
  (H_E) [E >= M (eps + rho) + (1 + n + ... + n^(k-1))].
[exp] does not appear: it only enters through (H_A) and (H_T).

Stdlib only, no mathcomp, so that the real arithmetic is not disturbed
by mathcomp's notations. *)

From Stdlib Require Import ZArith Reals Lra Lia.

Open Scope R_scope.

(** [sum_(i < k) f i], by recursion on [k]. *)
Fixpoint sumR (k : nat) (f : nat -> R) : R :=
  match k with 0 => 0 | S k' => sumR k' f + f k' end.

(** The integer polynomial [sum_(i < k) A_i j^i]. *)
Fixpoint Pz (A : nat -> Z) (k j : nat) : Z :=
  match k with

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| 0 => 0%Z
| S k' => (Pz A k' j + A k' * Z.of_nat j ^ Z.of_nat k')%Z
end.

(** Extensionality of [sumR] below [k]. *)
Lemma sumR_ext (k : nat) (f g : nat -> R) :
  (forall i, (i < k)%nat -> f i = g i) -> sumR k f = sumR k g.
Proof.
induction k as [|k IH]; intros H; simpl; [reflexivity|].
rewrite IH by (intros i Hi; apply H; lia).
rewrite H by lia; reflexivity.
Qed.

(** [sumR] is additive. *)
Lemma sumR_plus (k : nat) (f g : nat -> R) :
  sumR k (fun i => f i + g i) = sumR k f + sumR k g.
Proof. induction k as [|k IH]; simpl; [lra|rewrite IH; lra]. Qed.

(** [sumR] commutes with a scalar factor. *)
Lemma sumR_scal (k : nat) (c : R) (f : nat -> R) :
  sumR k (fun i => c * f i) = c * sumR k f.
Proof. induction k as [|k IH]; simpl; [ring|rewrite IH; ring]. Qed.

(** The triangle inequality for [sumR]. *)
Lemma Rabs_sumR_le (k : nat) (f g : nat -> R) :
  (forall i, (i < k)%nat -> Rabs (f i) <= g i) ->
  Rabs (sumR k f) <= sumR k g.
Proof.
induction k as [|k IH]; intros H; simpl.
- rewrite Rabs_R0; lra.
- apply Rle_trans with (Rabs (sumR k f) + Rabs (f k)); [apply Rabs_triangular|].
  apply Rplus_le_compat; [apply IH; intros i Hi; apply H; lia|apply H; lia].
Qed.

(** The integer polynomial, read in the reals. *)
Lemma IZR_Pz (A : nat -> Z) (k j : nat) :
  IZR (Pz A k j) = sumR k (fun i => IZR (A i) * INR j ^ i).
Proof.
induction k as [|k IH]; simpl; [reflexivity|].
rewrite plus_ISZR, mult_ISZR, IH, <- pow_ISZR, <- INR_ISZR_INZ.
reflexivity.
Qed.

Lemma real_lemma (k n j : nat) (M E : Z) (a : nat -> R) (A : nat -> Z)
  (rho eps y : R) :
  (0 < M)%Z -> (j <= n)%nat ->
  (forall i, (i < k)%nat -> Rabs (IZR (A i) - IZR M * frac_part (a i)) < 1) ->
  Rabs (y - sumR k (fun i => a i * INR j ^ i)) <= rho ->
  IZR M * (eps + rho) + sumR k (fun i => INR n ^ i) <= IZR E ->
  (exists z : Z, Rabs (y - IZR z) < eps) ->
  exists w : Z, (Z.abs (Pz A k j - M * w) <= E)%Z.
Proof.
intros HM Hj n HA HT HE [z Hz].
set (K := Pz (fun i => Int_part (a i)) k j).
set (S := sumR k (fun i => a i * INR j ^ i)).
set (D := sumR k (fun i =>

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      (IZR (A i) - IZR M * frac_part (a i)) * INR j ^ i)).
assert (HP : IZR (Pz A k j) = D + IZR M * S - IZR M * IZR K).
{ unfold D, S, K; rewrite !IZR_Pz, <- !sumR_scal.
  transitivity (sumR k (fun i =>
    ((IZR (A i) - IZR M * frac_part (a i)) * INR j ^ i
      + IZR M * (a i * INR j ^ i))
    + (-1) * (IZR M * (IZR (Int_part (a i)) * INR j ^ i))))).
  { apply sumR_ext; intros i _; unfold frac_part; ring. }
  rewrite sumR_plus, sumR_scal, sumR_plus; ring. }
assert (HD : Rabs D <= sumR k (fun i => INR n ^ i)).
{ apply Rabs_sumR_le; intros i Hi.
  rewrite Rabs_mult, (Rabs_right (INR j ^ i))
    by (apply Rle_ge, pow_le, pos_INR).
  apply Rle_trans with (1 * INR j ^ i).
  - apply Rmult_le_compat_r; [apply pow_le, pos_INR|].
    apply Rlt_le, HA; exact Hi.
  - rewrite Rmult_1_l; apply pow_incr; split; [apply pos_INR|].
    apply le_INR; exact Hjn. }
exists (z - K)%Z.
apply le_IZR; rewrite abs_IZR.
rewrite minus_IZR, mult_IZR, minus_IZR, HP.
assert (HM' : 0 < IZR M) by (apply IZR_lt; exact HM).
replace (D + IZR M * S - IZR M * IZR K - IZR M * (IZR z - IZR K))
  with (D - IZR M * (y - S) + IZR M * (y - IZR z)) by ring.
apply Rle_trans
  with (Rabs D + IZR M * Rabs (y - S) + IZR M * Rabs (y - IZR z)).
{ apply Rle_trans
  with (Rabs (D - IZR M * (y - S)) + Rabs (IZR M * (y - IZR z)));
  [apply Rabs_triangler|].
  rewrite Rabs_mult, (Rabs_right (IZR M)) by lra.
  apply Rplus_le_compat_r.
  unfold Rminus at 1; eapply Rle_trans; [apply Rabs_triangler|].
  rewrite Rabs_Ropp, Rabs_mult, (Rabs_right (IZR M)) by lra; lra. }
assert (IZR M * Rabs (y - S) <= IZR M * rho)
  by (apply Rmult_le_compat_l; [lra|exact HT]).
assert (IZR M * Rabs (y - IZR z) <= IZR M * eps)
  by (apply Rmult_le_compat_l; lra).
lra.
Qed.

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## C Annex: TaylorLink.v

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(** * Zimmermann's search (doc/htr.md): linking the reals to the scan

[TaylorReal.real_lemma] ends on [Pz], a sum over [Z], and
[TaylorScan.scan_complete] starts on [hpoly], a Horner form over
[int]. This file proves they are the same polynomial, and restates
[scan_complete] over [Z], in the shape [real_lemma] produces. *)

From Stdlib Require Import ZArith Reals Lia.
From APaulRocq Require Import TaylorReal.
From mathcomp Require Import all_ssreflect all_algebra ssrZ zify.
From APaulRocq Require Import Shift TaylorScan.

Set Implicit Arguments.

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Unset Strict Implicit.
Unset Printing Implicit Defensive.

Import GRing.Theory Num.Theory.

(** One more term of [Pz], without letting [/=] unfold the power. *)
Lemma PzE (f : nat -> Z) k j :
  Pz f k.+1 j =
    Z.add (Pz f k j) (Z.mul (f k) (Z.pow (Z.of_nat j) (Z.of_nat k))).
Proof. by []. Qed.

(** Peeling the constant term off [Pz]: the Horner step, over [Z]. *)
Lemma PzS (f : nat -> Z) k j :
  Pz f k.+1 j = Z.add (f 0) (Z.mul (Z.of_nat j) (Pz (fun i => f i.+1) k j)).
Proof.
elim: k => [|k IH]; first by rewrite !PzE /=; lia.
by rewrite PzE IH PzE Nat2Z.inj_succ Z.pow_succ_r; lia.
Qed.

(** [hpoly] and [Pz] compute the same integer. *)
Lemma hpoly_Pz (A : seq int) (j : nat) :
  Z_of_int (hpoly A j) = Pz (fun i => Z_of_int (nth 0%R A i)) (size A) j.
Proof.
elim: A => [|a A IH].
by rewrite [size _] /= PzS /= -IH; lia.
Qed.

(** [scan_complete] over [Z], with the polynomial written as [Pz]. *)
Theorem scan_completeZ (A : seq int) (M E : int) (n j : nat) (w : Z) :
  Z.lt (Z.mul 2 (Z_of_int E)) (Z_of_int M) -> (j < n)%N ->
  Z.le (Z.abs (Z.sub (Pz (fun i => Z_of_int (nth 0%R A i)) (size A) j)
    (Z.mul (Z_of_int M) w)))
    (Z_of_int E) ->
  j \in scan A M E n.
Proof.
rewrite -hpoly_Pz => EM jn Pw.
by apply: (@scan_complete A M E n j (int_of_Z w) _ jn); rewrite /P; lia.
Qed.

(** The search misses nothing. [y] stands for  $\frac{\exp(x_0 + j u)}{v}$ , [a i]
for the Taylor coefficient  $\frac{\exp(x_0) u^i}{(i! v)}$ , [eps] for  $2^{-m}$ .
Under the three hypotheses of the note, (H_A) on the [A_i], (H_T) the
Taylor bound [rho], and (H_E) on [E], if [y] is within [eps] of an
integer then [j] is a candidate. Real numbers are written with prefix
functions ([Rle], [Rplus], ...), since mathcomp owns the infix ones. *)
Theorem scan_exp (A : seq int) (M E : int) (n j : nat) (a : nat -> R)
  (rho eps y : R) :
  Z.lt 0 (Z_of_int M) -> Z.lt (Z.mul 2 (Z_of_int E)) (Z_of_int M) ->
  (j < n)%N ->
  (forall i, Peano.lt i (size A) ->
    Rlt (Rabs (Rminus (IZR (Z_of_int (nth 0%R A i)))
      (Rmult (IZR (Z_of_int M)) (frac_part (a i)))))) 1) ->
  Rle (Rabs (Rminus y (sumR (size A) (fun i => Rmult (a i) (pow (INR j) i))))))
    rho ->
  Rle (Rplus (Rmult (IZR (Z_of_int M)) (Rplus eps rho))
    (sumR (size A) (fun i => pow (INR n) i)))

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      (IZR (Z_of_int E)) ->
      (exists z : Z, Rlt (Rabs (Rminus y (IZR z))) eps) ->
      j \in scan A M E n.
Proof.
move=> M0 EM jn HA HT HE Hz.
have jn' : Peano.le j n by apply/leP; exact: ltnW.
have [w Hw] := real_lemma _ _ _ _ _ _ _ _ M0 jn' HA HT HE Hz.
exact: scan_completeZ EM jn Hw.
Qed.

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